

Project Title: Smart Legal Assistant for Indian Court Cases

What is this project?

This project is an AI-powered system designed to help lawyers and common citizens understand long, complex Indian court judgments. Instead of reading dozens of pages of legal text, a user can upload a document and get a **short, human-like summary** and **instant answers** to specific questions about the case.

How does it work?

The system uses "State-of-the-Art" Artificial Intelligence called **Large Language Models (LLMs)** from Hugging Face.

- **The Brain:** We take a powerful pre-trained model (like BART or T5) that already understands English and "fine-tune" it specifically on Indian legal documents. This teaches the AI to understand legal terms like "Petitioner," "Affidavit," and "Jurisdiction."
- **The Process:** 1. **Summarization:** The AI reads the entire judgment and writes a concise summary of the key facts and the final verdict.
2. **Q&A:** The user can ask questions like "*Why was the appeal rejected?*" and the AI finds the exact reason within the text.

Why is it better than before?

Previously, we used a system called **BiLSTM**. While good, it often struggled with very long documents and complex legal logic. By upgrading to **LLMs (Transformers)**, the system now:

- Understands the deep **context** of sentences better.
- Generates **smoother, more natural** summaries (Abstractive Summarization).
- Handles **longer documents** with much higher accuracy.

The Impact

By using this tool, legal professionals can save hours of manual reading, reducing the backlog of work and making legal information more accessible to everyone.